

ALEX YAN

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EDUCATION

Purdue University

B.S. Computer Science

West Lafayette, IN

May 2028

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Systems Programming, Computer Architecture, Programming in C, Discrete Mathematics, Linear Algebra, Introduction to Statistics

EXPERIENCE

Shopify

Software Engineering Intern

New York, NY

Sept. 2026 - Dec. 2026

- Incoming Fall 2026 Software Engineering Intern at Shopify

PNC Bank

Software Engineering Intern

Pittsburgh, PA

Jun. 2026 – Aug. 2026

- Delivered **5** Spring Boot validation features for loan origination workflows using Java, Gradle, MongoDB, and REST APIs, deploying through Red Hat OpenShift across testing environments.
- Building comprehensive JUnit test coverage for validation logic, verifying valid, invalid, and edge-case scenarios end-to-end with Postman against live MongoDB-backed configuration.
- Diagnosing and resolving intermittent Jenkins CI/CD failures caused by race conditions in asynchronous test execution, restoring reliable Gradle builds and test stability before merge.

Purdue Stack

Full Stack Developer

West Lafayette, IN

Sept. 2025 – May 2026

- Shipped a research collaboration platform in Next.js, FastAPI, and PostgreSQL for Purdue's Statistics Department, featuring role-based dashboards, real-time messaging, and JWT authentication. (Spring 2026)
- Delivered a TA assignment optimization system for Purdue's Mathematics Department serving 150–200 TAs per semester, reducing scheduling setup time by **25%** via a preference-weighted greedy matching algorithm in Python. (Fall 2025)

Rooh (Berkeley SkyDeck Pad-13 Incubator)

Software Engineering Intern

Berkeley, CA (Remote)

Jun. 2025 – Dec. 2025

- Engineered a real-time conversational AI app in React and Next.js, orchestrating an end-to-end voice pipeline with Google Cloud STT, LLM dialogue generation, and Google Cloud TTS, reducing response latency by **20%**.
- Architected Next.js backend data pipelines supporting a live AI simulator and async review tool, handling real-time audio streaming, buffering, and transcription at scale.
- Used React to redesign the marketing landing page with a dynamic scrollytelling component and a custom analytics dashboard generating session-based delivery and tone metrics.

PROJECTS

PyroTech | *Next.js, TypeScript, Google Gemini, WebSocket* | [Github](#)

Apr. 2026

- Developed an AI wildfire incident command simulation platform with a Google Gemini multi-agent orchestration layer coordinating 7 specialized agents across disaster, evacuation, resource, and infrastructure domains.
- Engineered a physics-based fire spread engine running at a **500ms tick rate** with dozer barrier mechanics and suppression zone modeling, streamed live to clients via WebSocket.
- Pulled real-world map data to power layered geospatial visualizations, supporting branching what-if scenario analysis for incident commanders.

Iris | *JavaScript, Chrome MV3, AWS Lambda, OpenAI* | [Github](#)

Feb. 2026

- Developed a voice-driven productivity assistant for Google Workspace at Carnegie Mellon's 24-hour flagship hackathon, targeting routine email, calendar, and document workflows across Gmail, Calendar, and Docs.
- Designed a serverless AWS Lambda backend and Chrome MV3 extension with wake-word detection, real-time audio capture, and LLM-driven intent planning.

Access Prep | *React, TypeScript, Flask, LangChain, Gemini* | [GitHub](#)

Jan. 2026

- Won **3rd Place** at the Capital One Tech Summit Hackathon by building a bilingual document preparation platform serving underrepresented populations across 5 immigration and financial aid pathways.
- Implemented a Flask REST API handling session management, file uploads, and AI validation, integrating LangChain and Gemini to power a context-aware AI chatbot guiding users through the platform.

SKILLS

Languages: Python, GoLang, Java, JavaScript, TypeScript, C, C++, SQL, Bash

Frameworks & Tools: React, Next.js, Flask, Node.js, Spring Boot, Tailwind CSS, Bitbucket, Github, Postman, Jenkins

Databases/Cloud: PostgreSQL, MongoDB, AWS (Lambda, Gateway), Google Cloud, Docker

AI Dev Tools: Claude Code, GitHub Copilot, Cursor, OpenAI Codex